



PingTool



About the Project

PingTool is a lightweight Bash command-line utility built for basic network reconnaissance. The user provides a target IP address or domain, and PingTool checks whether the host is reachable, scans it for commonly open ports, and saves every result to a timestamped report file. It's a small tool, but it reflects a real, everyday step in penetration testing workflows: confirm a host is alive, then see what's open on it.



Features

- Host discovery — pings a target 3 times and reports UP / NO RESPONSE along with the average round-trip time (RTT).
 - Port scanning — checks 10 commonly used ports (FTP, SSH, Telnet, SMTP, DNS, HTTP, HTTPS, SMB, MySQL, RDP) using Bash's built-in `/dev/tcp` pseudo-device, no external scanner required.
 - Service name mapping — each port is matched to its service name and printed in a clean, aligned table.
 - Report saving — automatically creates a `results/` folder and appends (never overwrites) a dated report for every scan.
 - Interactive menu — a simple numbered menu loop (1-3, plus 0 to exit) so the tool can be used without memorizing flags.
 - Flexible input — works with or without a command-line argument: `./pingtool.sh` or `./pingtool.sh 192.168.1.1`.
 - Input validation — target names are checked against a safe character pattern before being used, to prevent command injection through the `/dev/tcp` calls.
 - Environment checks — verifies that `bash` and `ping` are actually available before running, with a clear message if not.
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Installation

PingTool only needs Bash and standard Linux networking tools — there is nothing to compile and no external dependencies to install beyond what most Linux systems already ship with.

Requirements

- Bash 4+ (with /dev/tcp support enabled, which is the default on almost all Linux distributions)
- ping (part of iputils-ping)

Steps

```
# 1. Get the script onto your machine
git clone <your-repo-url>
cd pingtool

# 2. Make it executable
chmod +x pingtool.sh

# 3. (Optional) install ping if it's missing
sudo apt install iputils-ping
```

Usage

Run the script with or without a target:

```
./pingtool.sh
./pingtool.sh 192.168.1.1
```

You'll be dropped into an interactive menu:

```
|| PINGTOOL ||
```

```
[1] Check host (ping)
[2] Scan ports
[3] Save report
[0] Exit
```

Choose 1 to check whether a host is alive, 2 to scan its ports, and 3 to save everything you've found so far into a report file such as `results/scan_192.168.1.1_2026-07-08.txt`. Choose 0 at any time to exit.

 License

This project was created as an educational exercise for the Module 7 (Bash Scripting) exam at HAAD Training Center. It is free to use, modify, and share for learning purposes. No warranty is provided — use it responsibly and only against hosts and networks you own or have explicit permission to test.